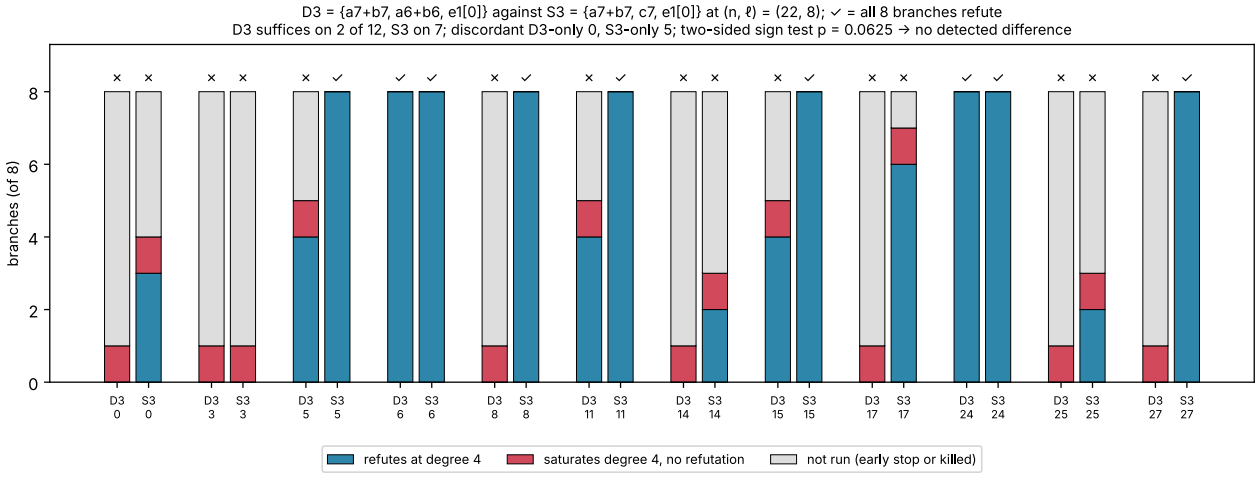
A second invariant guess on the pair does not replace the raw guess on x_3

Study `research/dreg_invariant_pair_guess_20261029/`, pre-registered. **Class:** stage diagnostic. **Registered verdict:** no detected difference.

Question

S3 = {a_7+b_7, c_7, e_1[0]} beat the raw guess at (22, 8) (2883). Does D3 = {a_7+b_7, a_6+b_6, e_1[0]}, which replaces the raw c_7 with a second invariant coordinate of the pair, do better?

Result



- D3 suffices on 2 of 12 draws and S3 on 7 of 12.
- All 5 discordant draws favour S3. The two-sided exact sign test gives p = 0.0625, so the registered verdict is no detected difference.
- D3 is never better on any draw. Where both fail, D3 fails no later.
- Controls valid: the instrument identity, the identity and exactness checks, and every draw decided.

Reading

- One invariant guess per summand pair is the useful part. The raw guess on the third summand carries weight that a second pair guess does not.
- The best measured guess rule stays S3′s.
- No row moves and none beats rho.